

MindEase

Mental Health Chatbot

Stress Detection & Psychological Support System

Project Type	Semester End Project — 4th Semester
Department	BS Computer Science
University	The Superior University, Lahore
Algorithms	Linear Regression Decision Tree SVM
Year	2025

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1. Abstract

MindEase is an AI-powered Mental Health Chatbot designed to detect psychological stress from natural language text and respond with empathetic, evidence-based support. Unlike conventional rule-based chatbots, MindEase employs a tri-algorithm machine learning pipeline — Linear Regression for continuous stress scoring, Decision Tree for categorical stress classification, and Support Vector Machine (SVM) for binary stress detection — all grounded in real-world psychology research. The system is built to serve all age groups, remain non-judgmental, and provide actionable relief strategies rooted in clinical psychology. This document presents the complete technical design, algorithm implementations, dataset details, UI/UX vision, and implementation roadmap for MindEase.

2. Introduction

2.1 Problem Statement

Mental health disorders — particularly stress and anxiety — affect over 970 million people globally (WHO, 2022). In Pakistan, awareness and access to professional mental health care remain critically low. Students, working professionals, and youth often lack a safe, private space to express what they are feeling without fear of judgment. Digital tools that bridge this gap are urgently needed.

2.2 Motivation & Real-World Impact

- 1 in 4 people will experience a mental health condition in their lifetime.
- Most people do not seek help due to stigma, cost, or lack of access.
- Early detection of stress patterns can prevent escalation to severe disorders.
- A chatbot available 24/7 can serve as a first point of support at zero cost.
- Combining ML with psychology creates a system that is both intelligent and empathetic.

3. System Overview

3.1 What MindEase Does

MindEase provides two entry points when a user opens the application:

Mode	Description	Best For
Question Mode	User answers a structured set of psychological questions. The system analyses responses and generates a stress profile.	Users unsure what to say or new to the platform.
Chat Mode	User types freely. The AI detects stress in real-time from conversational text and responds empathetically.	Users who want to express themselves naturally.

3.2 How MindEase Stands Out

Feature	Typical Chatbot	MindEase
Single algorithm	✓	✗ (3 algorithms, compared)
Psychology-backed responses	✗	✓ (CBT, mindfulness, breathing)
Non-judgmental design	✗	✓ (core design principle)
Stress score + category + binary	✗	✓
User session history tracking	✗	✓
All age groups supported	✗	✓
3D body pain mapping (vision)	✗	✓ (future feature)

4. Architecture & Flow

4.1 System Architecture

The system is divided into five logical layers, each handling a specific responsibility:

Layer 1 — Input Layer	User types text (Chat Mode) or answers questions (Question Mode) via GUI.
Layer 2 — NLP & Feature Extraction	Raw text is converted into numerical features: negative word count, punctuation ratio, sentence length, capital letter ratio, and lexical diversity.
Layer 3 — ML Engine	Three algorithms process features simultaneously: LR gives a score (0–10), DT gives a category (Low/Medium/High), SVM gives a binary label (Stressed/Not Stressed).
Layer 4 — Psychology Response Engine	Based on combined model output, a psychology-backed response and coping strategy is selected from a structured response library.
Layer 5 — Output Layer	Response is displayed on screen. Session data is logged for history tracking.

4.2 User Journey — Step-by-Step Flow

Step 1	User opens MindEase app
Step 2	Chooses: Question Mode or Chat Mode
Step 3	Provides input (answers / free text)
Step 4	Feature extraction converts text to numbers
Step 5	All 3 ML models analyse features simultaneously
Step 6	Combined decision: Stress level determined
Step 7	Psychology engine selects appropriate response
Step 8	Response + coping strategy displayed to user
Step 9	Session saved to user history

5. Algorithms — Deep Dive

5.1 Linear Regression — Stress Score Prediction

Linear Regression models the relationship between input features (X) and a continuous output (y). In MindEase, it predicts a stress score between 0 and 10.

Core Equation:

$$y = m \cdot x + b$$

where y = predicted stress score, x = input features, m = weight (slope), b = bias (intercept)

Cost Function — Mean Squared Error (MSE):

$$J(m, b) = (1/n) \times \sum (y_i - (m \cdot x_i + b))^2$$

This measures how wrong our predictions are. Lower MSE = better model.

Gradient Descent — Update Rules:

$$\partial J / \partial m = -(2/n) \times \sum x_i \cdot (y_i - \hat{y}_i)$$

$$\partial J / \partial b = -(2/n) \times \sum (y_i - \hat{y}_i)$$

$$m := m - \alpha \cdot (\partial J / \partial m) \quad b := b - \alpha \cdot (\partial J / \partial b)$$

α (alpha) = learning rate. Too large → overshoots. Too small → slow convergence.

How Learning Rate Affects Training:

Learning Rate	Effect	Result
Too Large (e.g. 0.9)	Overshoots minimum	Model diverges ✗
Too Small (e.g. 0.0001)	Very slow updates	Takes too long ✗
Just Right (e.g. 0.01)	Steady convergence	Optimal model ✓

5.2 Decision Tree — Stress Category Classification

A Decision Tree learns a set of if-then-else rules from training data. Each internal node tests a feature, each branch represents a decision, and each leaf is a predicted class. In MindEase, the output is: Low Stress / Medium Stress / High Stress.

Example Decision Path:

Negative Words > 5 ?

■■■ YES → Sentence Length > 15 ?

■ ■■■ YES → HIGH STRESS ■

■ ■■■ NO → MEDIUM STRESS ■

■■■ NO → LOW STRESS ■

Key Advantage: The decision path is fully explainable — you can tell the user exactly why a certain stress level was detected.

5.3 SVM — Binary Stress Detection

Support Vector Machine (SVM) finds the optimal hyperplane that maximally separates two classes. It is particularly effective for text classification tasks. In MindEase, SVM gives a binary output: Stressed (1) or Not Stressed (0). SVM is generally the most accurate of the three for text-based tasks.

Core Concept:

- SVM maps text features into high-dimensional space.
- It finds the widest possible margin between 'Stressed' and 'Not Stressed' data points.
- New input is classified based on which side of the decision boundary it falls on.
- Kernel trick (RBF kernel) allows non-linear boundaries without changing the algorithm.

5.4 Algorithm Comparison

Algorithm	Output Type	Output Example	Best Strength	Limitation
Linear Regression	Continuous score	7.3 / 10	Quantifies stress intensity	Requires scaling
Decision Tree	Category	HIGH STRESS	Explainable rules	Can overfit
SVM	Binary label	STRESSED	Highest accuracy on text	Slower on large data

6. Feature Extraction — Text to Numbers

Machine learning algorithms cannot understand raw text — they only process numbers. Feature extraction is the bridge between human language and mathematical models.

Feature	How It Is Calculated	Why It Matters	Example Value
Negative Word Count	Count words in negative lexicon (sad, hopeless, overwhelmed...)	Primary stress indicator	4
Sentence Length	Total word count in input	Longer venting = higher stress	23
Exclamation Marks	Count of ! characters	Emotional intensity signal	3
Capital Letter Ratio	CAPS letters / total letters	Indicates emotional escalation	0.18
Question Mark Count	Count of ? characters	Confusion, seeking help signal	2
Lexical Diversity	Unique words / total words	Low diversity = rumination	0.61
Positive Word Count	Count words in positive lexicon	Inverse stress indicator	1

Example:

Input Text:

"I CANT DO THIS ANYMORE!! everything feels hopeless and I dont know what to do???"

Extracted Features:

Negative Words: 2 | Sentence Length: 18 | Exclamation Marks: 2 | Capital Ratio: 0.21 | Question Marks: 3

→ LR Score: 8.4 | DT Category: HIGH STRESS | SVM: STRESSED

7. Psychology Integration

7.1 Psychological Framework

MindEase is not just a classification tool — it is designed to be a psychologically informed companion. The response system is grounded in three validated frameworks:

Framework	What It Is	How MindEase Uses It
Cognitive Behavioural Therapy (CBT)	Identifies and challenges negative thought patterns	Responses gently reframe negative self-talk
Mindfulness-Based Stress Reduction	Present-moment awareness techniques	Breathing exercises, grounding prompts
Positive Psychology	Building on strengths rather than fixing deficits	Highlights user progress, affirms resilience

7.2 Response Design Principles

- Non-Judgmental:** The system never criticises the user. Every response validates the user's feelings first.
- Empathy First:** Responses always begin with acknowledgment before offering advice.
- Strength-Focused:** The chatbot reminds users of their inner resilience and past successes.
- Graduated Support:** Low stress → tips. Medium stress → exercises. High stress → professional help guidance.
- Honest & Warm:** The chatbot does not give false reassurances. It is honest, kind, and supportive.

Sample Response Map:

Stress Level	Score	Sample Response
Low	0 – 3	"Glad you're doing okay! Here's a quick mindfulness moment for you..."
Medium	4 – 6	"That sounds tough. Let's try the 4-7-8 breathing technique together..."
High	7 – 10	"You're not alone in this. What you're feeling is real. Please consider reaching out to someone you trust..."

8. Dataset

8.1 Source & Description

The project uses publicly available, ethically sourced datasets from Kaggle. Two recommended datasets are:

Dataset	Source	Labels	Size
Stress Analysis in Social Media	Kaggle	Stress / No Stress	~2,800 posts
Dreaddit (Reddit Stress Dataset)	Kaggle / GitHub	Stress / No Stress	~3,553 posts
SentimentHealth	Kaggle	Positive / Negative / Neutral	~5,000 posts

8.2 Sample Data Format

```
text, label, stress_score
"I cant sleep, everything feels heavy",1,8.2
"Had a great day with friends, feeling good",0,1.4
"Work is piling up and I feel overwhelmed",1,7.1
```

9. Implementation Plan

9.1 Week-by-Week Roadmap

Week 1 Setup & Data

- ✓ Set up Python environment (VS Code, libraries)
- ✓ Download dataset from Kaggle
- ✓ Explore and clean data (remove nulls, lowercase text)
- ✓ Build feature extraction module

Week 2 Linear Regression

- ✓ Implement LR from scratch (NumPy only)
- ✓ Train on stress dataset
- ✓ Test with sample inputs
- ✓ Plot regression line and cost curve

Week 3 Decision Tree + SVM

- ✓ Implement Decision Tree algorithm
- ✓ Add SVM (can use sklearn for this)
- ✓ Run all 3 models on same data
- ✓ Compare accuracy, precision, recall

Week 4 Chatbot Logic + GUI

- ✓ Build psychology response library
- ✓ Integrate models with response engine
- ✓ Build Tkinter GUI with Chat + Question mode
- ✓ Style GUI (colors, layout, fonts)

Week 5 Testing + Documentation

- ✓ Test with 10+ different user inputs
- ✓ Fix bugs and edge cases
- ✓ Complete full documentation
- ✓ Prepare demo / presentation

9.2 Tech Stack

Component	Technology	Purpose
Language	Python 3.10+	Core programming language
ML (LR)	NumPy (from scratch)	Linear Regression implementation

ML (DT + SVM)	NumPy / scikit-learn	Classification algorithms
NLP	NLTK / re (regex)	Text cleaning and feature extraction
GUI	Tkinter	Desktop application interface
Data	Pandas	Dataset loading and manipulation
Visualisation	Matplotlib	Regression plots, cost curves
Dataset	Kaggle CSV	Labeled stress text data

10. UI/UX Design Vision

MindEase is designed to feel calm, safe, and welcoming — never clinical or robotic. The visual language uses soft blush tones, rounded elements, and clear typography.

Colour Palette: Soft blush (#FDF0F3), dusty rose (#C0576A), warm cream (#FDF6F0) — colours associated with calm and warmth.

Typography: Clean sans-serif fonts. Larger font for responses. Friendly, readable at all ages.

Home Screen: Two clear cards: 'Answer Questions' and 'Start Chatting'. No clutter.

Chat Screen: Bubble-style conversation. User messages on right (blush), bot on left (white). Stress indicator badge visible.

Question Mode: One question at a time. Progress bar visible. No time pressure.

Results Screen: Stress score shown as a calming dial/circle. Category label. Immediate suggestion card.

History Screen: Timeline of past sessions. Trends shown in a simple line graph.

Future: 3D Body Map: Interactive human figure where user taps pain/tension locations. Maps to stress response suggestions.


```
return float(np.clip(score, 0, 10)) # clamp to 0-10 range
```

6. Test Cases

```
test_cases = [
    ([0.1, 0.3, 0.0, 0.02], 'I am doing okay today'),
    ([0.4, 0.6, 0.2, 0.10], 'Work is stressful lately'),
    ([0.8, 0.9, 0.8, 0.35], 'I CANT HANDLE THIS ANYMORE!!!'),
]

for features, text in test_cases:
    score = predict_stress(features)
    print(f'Text: {text[:40]}')
    print(f' Stress Score: {score:.2f} / 10')
```

12. Model Comparison & Results

After training all three models on the stress dataset, their performance is compared using standard evaluation metrics. This comparison is a key differentiator of MindEase — most student projects use only one model.

Model	Metric	Expected Result	Interpretation
Linear Regression	MSE / R ² Score	MSE < 1.5, R ² > 0.85	Strong correlation between features and stress score
Decision Tree	Accuracy	~82–88%	Correctly classifies 8 in 10 stress levels
SVM	Accuracy + F1	~88–93%	Best overall classifier for text stress detection

Key Insight:

SVM performs best on binary classification. Decision Tree is most interpretable. Linear Regression is unique in providing a continuous score rather than a category. MindEase uses all three outputs together for the most complete picture of user stress.

13. Future Scope

- **Voice Input:** Allow users to speak instead of type. Analyse vocal tone and speed as additional stress indicators.
- **3D Body Pain Mapping:** Interactive human figure on screen. User taps where they feel physical tension. System maps to stress-related somatic responses.
- **Multilingual Support:** Add Urdu, Punjabi, and Hindi language support to serve Pakistani users in their native languages.
- **Deep Learning Upgrade:** Replace SVM with a fine-tuned BERT model for significantly higher text understanding accuracy.
- **Mobile App:** Convert to Android/iOS application using Flutter or React Native for wider reach.
- **Professional Referral:** For high stress scores, automatically suggest nearby mental health professionals or helpline numbers.
- **Wearable Integration:** Connect with smartwatch heart rate and sleep data for biometric stress validation.
- **Research Publication:** Dataset, model comparisons, and findings can be compiled into an academic paper.

14. Conclusion

MindEase represents a meaningful convergence of machine learning and clinical psychology, designed not only as an academic exercise but as a foundation for a genuinely useful real-world application. By implementing three distinct algorithms — Linear Regression, Decision Tree, and SVM — and grounding the system's responses in validated psychological frameworks, MindEase offers a more complete and compassionate approach to digital mental health support than conventional chatbots.

The system's dual-mode interface (Question Mode and Chat Mode), non-judgmental design philosophy, and session history tracking make it accessible and valuable to users of all age groups. The documentation presented here provides a complete blueprint for both the academic submission and the continued development of MindEase as a deployable product.

Mental health matters. Technology, when designed with empathy and scientific rigour, can play a meaningful role in supporting it.

15. References

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